

Real-Time Health Monitoring System

Project Overview

This project is a real-time health monitoring system that predicts the risk of heart attack, diabetes (sugar level), fever, and blood pressure abnormalities (high or low). It is built using Python (Flask), trained ML models (Random Forest), and a responsive frontend using Bootstrap 5.

Technologies Used

- Python
- Flask
- Scikit-learn
- Pandas
- NumPy
- HTML, CSS (Bootstrap 5)
- joblib
- FPDF (for this report)

Model Predictions

1. Heart Attack Risk: Based on age, gender, systolic BP, cholesterol, and heart rate.
2. Sugar (Diabetes) Risk: Based on age, gender, and glucose level.
3. Fever: Detected based on temperature $> 100.4^{\circ}\text{F}$.
4. Blood Pressure: Categorized as Normal, High, or Low based on SBP/DBP.

File Structure

- app.py: Flask backend
- templates/: HTML frontend

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- models/: Trained ML models
- dataset/: Sample health data
- history.csv: Stores prediction history
- requirements.txt: Python dependencies
- docs/: Documentation (PDF, PPT)

Conclusion

This system helps in preliminary health analysis and awareness by predicting multiple health parameters in real-time. It can be extended with real sensor APIs for live patient monitoring.